# Blockchain Technology (BT)

Short theory, keywords and probable viva / exam questions for every practical

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# Practical 15: Installation of MetaMask and Spending Ether per Transaction

## Topics in short

**MetaMask:** Browser-extension and mobile crypto wallet for Ethereum and EVM chains. Stores keys locally and connects websites (dApps) to the blockchain.

**Installation:** Add the extension from the official site, create a wallet, set a password, back up the 12-word Secret Recovery Phrase, and switch to a test network (e.g. Sepolia).

**Gas:** Unit of computation on Ethereum. Fee = gas used x gas price. After EIP-1559: fee = gas used x (base fee + priority fee/tip). The base fee is burned.

**Gas limit:** Maximum gas the sender allows; a simple ETH transfer uses 21,000 gas.

## Keywords

MetaMask, wallet, Ethereum, Ether, Wei, Gwei, gas, gas limit, gas price, base fee, priority fee, EIP-1559, test network, Sepolia, faucet, secret recovery phrase

## Probable questions (with short answers)

**Q1. What is MetaMask?**

Ans: A non-custodial crypto wallet that lets users manage Ethereum accounts and interact with dApps.

**Q2. What is gas?**

Ans: The unit measuring computational work of a transaction or contract execution.

**Q3. How is the transaction fee calculated?**

Ans: Gas used x gas price (base fee + priority fee).

**Q4. Gas used by a simple ETH transfer?**

Ans: 21,000 gas.

**Q5. What is Gwei?**

Ans: 1 Gwei = 10^9 Wei = 10^-9 ETH; gas prices are quoted in Gwei.

**Q6. What happens if the gas limit is too low?**

Ans: The transaction fails (out of gas) and the gas used is still charged.

**Q7. What is a test network and a faucet?**

Ans: Testnet: a copy of Ethereum with free test Ether. Faucet: website that gives test Ether.

**Q8. What is the Secret Recovery Phrase?**

Ans: 12-word seed that can restore the wallet; anyone with it controls the funds.

**Q9. What is EIP-1559?**

Ans: Ethereum upgrade that introduced a burned base fee plus an optional tip.

**Q10. Why do gas prices change?**

Ans: They depend on network demand (congestion).

# Practical 16: Create Your Own Wallet using MetaMask

## Topics in short

**Crypto wallet:** Software or hardware that stores the private keys used to sign transactions. Coins stay on the blockchain; the wallet only holds keys.

**Keys and address:** Private key (secret, signs transactions) -> public key -> address (last 20 bytes of the Keccak-256 hash of the public key, shown as 0x...).

**Wallet types:** Hot (online: MetaMask, mobile apps) vs cold (offline: hardware, paper). Custodial (exchange holds keys) vs non-custodial (user holds keys).

**Transaction flow:** Enter receiver address and amount -> sign with private key -> broadcast -> miners/validators include it in a block -> confirmation.

## Keywords

cryptocurrency, wallet, private key, public key, address, digital signature, ECDSA, seed phrase, hot wallet, cold wallet, custodial, non-custodial, transaction hash, Etherscan, confirmation, nonce

## Probable questions (with short answers)

**Q1. Difference between private and public key?**

Ans: Private key signs transactions and must be secret; the public key/address is shared to receive funds.

**Q2. Hot vs cold wallet?**

Ans: Hot is connected to the internet (convenient, less secure); cold is offline (more secure).

**Q3. Custodial vs non-custodial wallet?**

Ans: Custodial: a third party holds your keys. Non-custodial: you hold them.

**Q4. How is an Ethereum address generated?**

Ans: Last 20 bytes of the Keccak-256 hash of the public key.

**Q5. What is a digital signature?**

Ans: Proof created with the private key that the owner authorized the transaction (ECDSA on secp256k1).

**Q6. What is a transaction hash?**

Ans: Unique ID of a transaction, used to look it up on Etherscan.

**Q7. What is a nonce in an Ethereum account?**

Ans: Counter of transactions sent from the account; prevents replay.

**Q8. What happens if you lose the seed phrase?**

Ans: The wallet cannot be recovered; funds are lost.

**Q9. Name some cryptocurrencies.**

Ans: Bitcoin, Ether, Solana, Litecoin, stablecoins like USDT.

**Q10. What is a block explorer?**

Ans: Website (e.g. Etherscan) to view blocks, transactions and addresses.

# Practical 17: Bank Account Smart Contract (Deposit, Withdraw, Balance)

## Topics in short

**Smart contract:** Program stored on the blockchain that runs automatically when called. Immutable once deployed; runs on the Ethereum Virtual Machine (EVM).

**Solidity:** Statically typed language for Ethereum contracts. Key parts: pragma, contract, state variables, functions, modifiers, events, mapping.

**Bank contract:** mapping(address => uint) balances; deposit() is payable and adds msg.value; withdraw(amount) checks require(balance >= amount), reduces balance, then transfers; getBalance() is a view function.

**Deployment:** Write in Remix IDE, compile, deploy on Remix VM or a testnet via MetaMask (Injected Provider).

## Keywords

smart contract, Solidity, EVM, Remix IDE, pragma, mapping, msg.sender, msg.value, payable, view, pure, require, modifier, event, revert, reentrancy, checks-effects-interactions, ABI, bytecode

## Probable questions (with short answers)

**Q1. What is a smart contract?**

Ans: Self-executing code on a blockchain that enforces rules without a middleman.

**Q2. What does payable mean?**

Ans: A function marked payable can receive Ether.

**Q3. msg.sender vs msg.value?**

Ans: msg.sender = address calling the function; msg.value = Ether sent with the call (in Wei).

**Q4. view vs pure functions?**

Ans: view reads state but does not change it; pure neither reads nor changes state. Both cost no gas when called externally.

**Q5. Use of require()?**

Ans: Checks a condition; if false, reverts the transaction and refunds remaining gas.

**Q6. What is a mapping?**

Ans: Key-value store, e.g. mapping(address => uint) for balances.

**Q7. What is a reentrancy attack and how to prevent it?**

Ans: Malicious contract re-calls withdraw before balance is updated. Update state before transferring (checks-effects-interactions) or use a reentrancy guard.

**Q8. What is the ABI?**

Ans: Application Binary Interface: JSON description of contract functions used to call it.

**Q9. What is the EVM?**

Ans: The Ethereum Virtual Machine that executes contract bytecode on every node.

**Q10. Can a deployed contract be changed?**

Ans: No, code is immutable; upgrades need proxy patterns or a new deployment.

# Practical 18: Student Data Smart Contract (Structures, Arrays, Fallback)

## Topics in short

**Structure:** struct Student { uint id; string name; uint marks; } groups related fields.

**Arrays:** Student[] students; push() adds a record, length gives the count. Dynamic or fixed size.

**Fallback and receive:** receive() runs when plain Ether is sent with no data; fallback() runs when no function matches or data is sent. Must be external.

**Gas and fee:** Writing to storage is the most expensive operation; observe gas used and the transaction fee in Remix/Etherscan after each call.

## Keywords

struct, array, dynamic array, push, storage, memory, calldata, fallback function, receive function, external, event, gas used, transaction fee, deployment cost

## Probable questions (with short answers)

**Q1. What is a struct in Solidity?**

Ans: A user-defined type grouping related variables.

**Q2. storage vs memory vs calldata?**

Ans: storage: permanent on-chain; memory: temporary during a call; calldata: read-only function input.

**Q3. What is a fallback function?**

Ans: Special function executed when no other function matches the call or data is sent.

**Q4. Difference between fallback() and receive()?**

Ans: receive() handles plain Ether transfers with empty data; fallback() handles everything else.

**Q5. Why is writing data costly?**

Ans: Every node stores it permanently; SSTORE is one of the costliest opcodes.

**Q6. Why does deployment cost more gas than a function call?**

Ans: Deployment stores the whole contract bytecode on the chain.

**Q7. Fixed vs dynamic array?**

Ans: Fixed has a set length (uint[5]); dynamic can grow with push().

**Q8. What is an event?**

Ans: A log emitted by a contract, cheap storage readable by off-chain apps.

**Q9. How to view gas used?**

Ans: In the Remix console transaction details or on Etherscan for testnet deployments.

**Q10. Visibility specifiers?**

Ans: public, private, internal, external.

# Practical 19: Survey Report on Types of Blockchains and Use Cases

## Topics in short

**Blockchain:** Distributed, append-only ledger of blocks linked by hashes. Each block holds transactions, timestamp, previous hash and nonce. Tamper-evident and decentralized.

**Public blockchain:** Anyone can join and validate (Bitcoin, Ethereum). Fully decentralized, slower.

**Private blockchain:** One organization controls access (Hyperledger Fabric). Fast, less decentralized.

**Consortium / Federated:** Group of organizations jointly control it (R3 Corda, banking consortia).

**Hybrid:** Mix of public and private: private data with public verification (Dragonchain, XinFin).

**Consensus:** PoW (mining), PoS (staking), PBFT, PoA. Use cases: cryptocurrency, supply chain, healthcare records, voting, land registry, identity, cross-border payments.

## Keywords

distributed ledger, decentralization, block, hash, Merkle tree, immutability, consensus, Proof of Work, Proof of Stake, PBFT, Proof of Authority, public, private, consortium, hybrid, permissioned, permissionless, Hyperledger, use cases

## Probable questions (with short answers)

**Q1. What is a blockchain?**

Ans: A decentralized ledger of hash-linked blocks shared across many nodes.

**Q2. Public vs private blockchain?**

Ans: Public: open, permissionless. Private: restricted, permissioned, controlled by one organization.

**Q3. What is a consortium blockchain?**

Ans: A permissioned chain governed by a group of organizations.

**Q4. What is consensus? Give examples.**

Ans: Agreement among nodes on the ledger state: PoW, PoS, PBFT, PoA.

**Q5. PoW vs PoS?**

Ans: PoW: miners solve puzzles using energy. PoS: validators stake coins; far less energy.

**Q6. Why is a blockchain immutable?**

Ans: Each block holds the previous block's hash; changing one breaks all later links.

**Q7. What is a Merkle tree?**

Ans: Tree of hashes summarizing all transactions in a block into one Merkle root.

**Q8. Give three real-world use cases.**

Ans: Supply-chain tracking, medical records, digital identity, voting, payments.

**Q9. Permissioned vs permissionless?**

Ans: Permissioned needs approval to join; permissionless is open to all.

**Q10. Limitations of blockchain?**

Ans: Scalability, energy use (PoW), irreversibility of mistakes, regulation.